



AEGIS

Anonymization & Exchange Gateway
for Imaging Studies



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A multi-cloud platform for **HIPAA-compliant de-identification and sharing**
of medical imaging data — for research teams and radiology departments
alike.

HIPAA COMPLIANT

3 CLOUDS LIVE

BAA AVAILABLE

ZERO INSTALL AT SENDING SITES

Matthew L. Senjem, M.S. · AEGIS Imaging LLC · March 4, 2026 · aegisimaging.ai

Sharing medical images safely is harder than it should be

De-identification tools frequently fail. A 2015 study tested 10 free DICOM de-identification tools. Only 1 of 10 removed all required PHI with default settings. Four tools achieved success rates of 26% or less. (*Aryanto et al., European Radiology, 2015*)

Metadata removal alone isn't enough for brain imaging. Face-recognition software matched de-identified brain MRI participants to photographs in **83% of cases** — using only scan geometry, zero metadata. (*Schwarz et al., NEJM, 2019*)

Burned-in PHI is invisible to tag-level tools. Patient names, dates, and accession numbers are routinely overlaid on image pixels. Tag-level tools miss this entirely. (*Vcelak et al., Int'l J Medical Informatics, 2019*)

Data sharing is now federally mandated — but tooling lags. The NIH Data Management and Sharing Policy (Jan 25, 2023) requires all NIH-funded investigators to share scientific data as a condition of their grant award. Institutions are obligated to share data they have no reliable, low-friction way to transmit. (*NIH NOT-OD-21-013*)

Installing software at sending sites is a barrier. Platforms like XNAT and MIRC CTP require desktop clients or server daemons at every participating hospital — months of IT negotiation per site.

The infrastructure gap in medical imaging

\$3.9B

Medical image exchange
market (2024)

\$9.9B

Projected market by 2034
(9.7% CAGR)

~62%

Imaging studies lacking
compliant sharing pipelines

Why now:

- **NIH DMS Policy (2023)** creates a compliance obligation for tens of thousands of active grants — federally funded research *must* share imaging data
- **MIDI-B benchmark (NCI, MICCAI 2024)** confirmed burned-in pixel PHI and free-text fields remain unsolved; no existing tool handles all cases reliably
- **Growing multi-site trial volume** in oncology (PSMA PET, amyloid PET) and neurology (tau PET, fMRI) demands scalable collection from distributed hospital networks
- **Cloud DICOM infrastructure** (GCS, S3, Azure Blob) is now mature commodity, reducing the barrier to building managed platforms

The result: Multi-site studies delayed by months. Contract execution alone averaged 7.9 months per US site in a 2021 global randomized trial analysis. Software installation adds further delay. AEGIS eliminates both.

Two-phase de-identification — addressing every known gap

Phase 1 — In the browser (before upload):

- └─ DICOM tags stripped per PS3.15 Annex E Basic Profile
- └─ 18 HIPAA Safe Harbor identifiers addressed
- └─ Before-and-after tag diff preview — reviewer confirms before sending
 - └─ Only de-identified data is transmitted — PHI never leaves the hospital

↓

Phase 2 — On the server (after upload):

- └─ OCR scan of image pixels for burned-in text
- └─ Head imaging → automated defacing (facial surface removal)
- └─ Protocol compliance, QC, BIDS conversion, metadata classification
 - └─ Administrator review and approval — nothing shared without human sign-off

Key principles:

- No software to install at sending sites — any browser, any OS, zero IT approval
- All modalities: MRI, CT, PET, PET/CT, ultrasound, X-ray, mammography, RT, nuclear medicine
- Automated defacing is reviewable — side-by-side before/after in the admin dashboard
- Every action is logged — upload, approval, rejection, export — timestamped and actor-attributed

7 automated processing services — dispatched in dependency order

Service	What It Does
Classification	Reads DICOM headers to fill in modality and body part; re-evaluates routing rules after classification
PHI Detection	OCR scans image pixels for burned-in text (names, dates, accession numbers) that tag tools miss
Protocol Compliance	Verifies acquisition parameters (TR, TE, flip angle, resolution) against per-scanner templates
Defacing	Removes facial surface geometry from head MRI/CT/PET volumes; multiple backends with automatic fallback
QC	5 automated checks: file integrity, slice consistency, SNR estimation, coverage completeness, missing slices
BIDS Conversion	Converts DICOM to NIfTI with BIDS-compliant structure and JSON sidecar metadata (dcm2niix)
Synthetic MRI	Generates synthetic brain MRI phantoms for pipeline testing and defacing demos

AI-native operations: An MCP server exposes 191 typed tools (96 read + 95 write) to AI agents — Claude, Cursor, or any MCP-compatible agent can query studies, trigger pipeline steps, and triage stuck studies autonomously.

Auto-pipeline: After routing rules evaluate, services dispatch in the correct dependency order automatically. No manual clicks required.

One platform, two markets

	Research Teams	Radiology Departments
Entry point	Browser upload link sent to external sites	DIMSE C-STORE receive / batch import from PACS
Key need	NIH DMS compliance, multi-site consistency	HIPAA de-identification, complete audit trail
Core differentiator	Zero install at sending sites	No PACS/VNA integration or replacement
Pipeline features	Defacing, protocol compliance, QC, BIDS	Burned-in PHI detection, routing rules, export shares
Review workflow	PI/coordinator approves before archive	Radiologist/admin approves before release

Research teams get a browser link they can send to any participating site — no IT negotiation per hospital. The research pipeline (protocol compliance, automated QC, BIDS conversion, defacing) ensures data arrives analysis-ready for NIH-funded multi-site studies.

Radiology departments get a centralized de-identification gateway with a complete audit trail, burned-in PHI detection, and configurable routing rules — without replacing their PACS or embarking on a separate enterprise integration project.

Enterprise adoption path — no rip-and-replace required

AEGIS works alongside existing infrastructure. Studies arrive via DIMSE C-STORE push from your PACS, or batch-imported from network storage. The de-identification engine, routing rules, and audit trail sit between your PACS and any downstream consumer.

Concern	AEGIS Answer
PACS integration	No PACS/VNA replacement — receives via standard DIMSE C-STORE (port 11112) or DICOMweb STOW-RS
Software at sending sites	None — external sites use a browser link; no desktop app, no IT approval at participating hospitals
Cloud preference	GCP, AWS, or Azure in your existing cloud tenancy; or Docker Compose for evaluation
Compliance readiness	BAA available; HIPAA audit trail from day one; SOC 2 Type I audit in progress
Time to first study	Docker Compose evaluation in minutes; cloud pilot in days

Immediate internal use cases:

- Multi-site clinical trial imaging data collection — send a browser link to each site
- AI/ML training data preparation — de-identify and QC at scale before model training
- Real-world evidence databases — build a HIPAA-compliant archive from your clinical store
- Cross-department and cross-institution imaging data sharing with full audit trail

Pilot structure: A project-isolated instance can be deployed in your cloud tenancy within days. You control who uploads and who accesses the admin dashboard. No data leaves your tenancy.

Live today — three clouds, production-grade

7

days to GCP production

3

clouds live (GCP · AWS · Azure)

2,100+

automated tests

191

MCP AI agent tools

What's Live	Details
GCP	14 Cloud Run services + DIMSE VM · <code>api.aegisimaging.ai</code> · <code>admin.aegisimaging.ai</code>
AWS	14 ECS Fargate services + EC2 DIMSE · <code>aws.api.aegisimaging.ai</code>
Azure	14 Container Apps + DIMSE VM · <code>azure.api.aegisimaging.ai</code>
Cross-cloud routing	GCP→AWS→Azure STOW-RS verified live; bidirectional API key auth
CI/CD	Single merge to develop auto-deploys all 15 services on all 3 clouds simultaneously
API	322+ routes, 17-tab admin dashboard, full RBAC, audit trail, webhook subscriptions
Private beta	Invite-gated; research institutions and imaging centers welcome

Development velocity: 1,300+ git commits, built entirely with AI-assisted tooling on a single-engineer team.

Key differentiators from existing platforms

	AEGIS	ENCOG (Enlitic)	XNAT	Flywheel	MIRC CTP
Install at sending site	None — browser	PACS/VNA integration	Desktop Java client	CLI tool	Java application
Hosting	GCP · AWS · Azure	On-prem / hybrid	Self-hosted	Commercial SaaS	On-premises
Burned-in PHI detection	✓ OCR	✓ AI CV	×	✓	×
Automated defacing	✓ Reviewable	×	Manual / plugin	✓	×
Research pipeline	✓ Protocol / QC / BIDS	×	✓	✓	×
DIMSE C-STORE receive	✓	✓	✓	✓	✓
All DICOM modalities	✓	4 listed	Neuroimaging	✓	✓
AI-native operations (MCP)	✓ 191 tools	×	×	×	×

Five key differences:

1. **No software at sending sites** — full anonymization in the browser; ENCOG, XNAT, and MIRC CTP all require software at the sending institution
2. **Two-phase de-identification** — tag-level (browser) + pixel-level OCR (server) in a unified pipeline
3. **Automated, reviewable defacing** — not optional or absent; side-by-side before/after in the dashboard
4. **Dual-market platform** — research features (protocol compliance, QC, BIDS) + enterprise features (DIMSE, routing, audit trail) extend a shared core

5. Multi-cloud without re-architecture — same codebase, same API, same frontends on GCP, AWS, and Azure

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From first commit to three-cloud production — and a clear path to enterprise

Milestone	Date	Status
1 · Foundation + GCP Production — Go API, 7 Python microservices, DIMSE SCP, MCP server, admin dashboard, Terraform IaC, CI/CD	Feb 17–24, 2026	✓ Live
2 · Private Beta — Invite-gated early adopters; BAA finalized; SOC 2 Type I audit initiated; DIMSE C-MOVE / C-FIND	Q1 2026	● Active
3 · AWS Deployment — ECS Fargate, RDS, S3, ALB + Cognito; cross-cloud GCP→AWS routing verified	Feb 25, 2026	✓ Live
4 · Azure Deployment — Container Apps, PostgreSQL Flex, Azure Blob; GitHub Actions OIDC CI/CD	Feb 26, 2026	✓ Live
5 · Beta Hardening + Compliance — Production email, BAA countersigning, SOC 2 Type I audit, DIMSE C-MOVE, AWS Marketplace listing	Q2 2026	→ Planned
6 · SOC 2 Type II + Enterprise Integrations — HL7 FHIR notifications, DICOM conformance statement v2, cross-tenant federated sharing	Q3 2026	→ Planned
7 · Enterprise GA — SOC 2 Type II report, on-premises Helm chart, multi-tenant SaaS, PACS/VNA native query-retrieve	Q4 2026	→ Planned

All four completed milestones were delivered by a single engineer using AI-assisted development tooling. Three clouds live in 9 days from first commit.

DISCUSSION

Next steps — what we're exploring

The opportunity we're presenting:

AEGIS is production-ready across three major clouds and actively accepting pilot institutions. We are exploring internal partnerships and backing to accelerate the path from private beta to enterprise GA.

Three ways to engage:

Path	Description
Internal pilot deployment	Deploy a project-isolated AEGIS instance in your cloud tenancy — evaluate the full pipeline with your own imaging data under your BAA. No data leaves your environment.
Research collaboration	Use AEGIS to collect and de-identify imaging data for an active multi-site study — NIH DMS compliant, protocol QC and BIDS export included.
Backing / partnership	Support continued development toward SOC 2 Type II, BAA finalization, DIMSE C-MOVE, and AWS Marketplace listing — in exchange for early-adopter positioning and co-development input on features.

What happens next (Q2 2026):

- Production email configuration (share notifications, pipeline alerts)
- Business Associate Agreement countersigning workflow
- SOC 2 Type I audit initiation — engages auditor, starts 6-month Type II observation window
- DIMSE C-MOVE / C-FIND for active PACS pull
- AWS Marketplace listing for enterprise procurement

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